

Manufacturing Defect Detection Using CNN

Aim

To develop a Convolutional Neural Network (CNN) model for automatically detecting and classifying manufacturing defects from steel surface images using the NEU Surface Defect Dataset.

Objective

The objective of this project is to build a deep learning-based image classification system that can identify different types of surface defects in steel products.

Dataset Used

NEU Surface Defect Database

Classes:

1. Crazing
2. Inclusion
3. Patches
4. Pitted Surface
5. Rolled-in Scale
6. Scratches

Libraries Used

TensorFlow, NumPy, Scikit-Learn

Methodology

1. Data Loading
2. Data Preprocessing
3. Model Building
4. Model Training
5. Model Evaluation

CNN Architecture

Input -> Conv2D(32) -> MaxPooling -> Conv2D(64) -> MaxPooling -> Conv2D(128) -> MaxPooling -> Flatten -> Dense(128) -> Dropout(0.5) -> Softmax Output

Training Results

Epoch 1: Accuracy=24.48%, Validation Accuracy=18.40%
Epoch 2: Accuracy=49.65%, Validation Accuracy=83.68%
Epoch 3: Accuracy=65.80%, Validation Accuracy=64.58%

Final Evaluation

Validation Accuracy = 61.81%

Classification Report

Crazing: Precision=1.00 Recall=0.77 F1=0.87

Inclusion: Precision=0.74 Recall=0.71 F1=0.72

Patches: Precision=0.98 Recall=0.92 F1=0.95

Pitted Surface: Precision=0.33 Recall=0.90 F1=0.48

Rolled-in Scale: Precision=0.00 Recall=0.00 F1=0.00

Scratches: Precision=1.00 Recall=0.40 F1=0.57

Output Screenshots

Insert Dataset Structure Screenshot Here

Insert Model Summary Screenshot Here

Insert Training Output Screenshot Here

Insert Classification Report Screenshot Here

Observations

1. CNN successfully learned defect patterns.
2. Patches and Crazing achieved high accuracy.
3. Rolled-in Scale was difficult to classify.
4. More epochs may improve performance.
5. Data augmentation improved robustness.

Advantages

Automated quality inspection, reduced human effort, faster defect identification and consistent quality control.

Applications

Steel manufacturing, quality assurance systems, automated inspection lines and industrial defect monitoring.

Conclusion

The CNN model achieved 61.81% validation accuracy and demonstrated the effectiveness of deep learning for automated manufacturing defect detection.